DEVOPS CAPSTONE PROJECT

Helpdesk

Automated End-to-End Delivery, Security & Monitoring Pipeline

CANDIDATE
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REPOSITORY
github.com/Yashwardhan-ed/DevOps-Project

TECH STACK
FastAPI • React 18 • Docker • GitHub Actions • Terraform • EKS • Helm • Prometheus

MODULE 1 & 2 • WHAT I BUILT

Application Domain & Core Scope

Helpdesk Support Platform

- **Core Functionality:**
  Operations ticket management system enabling engineering and IT teams to log, triage, assign, and resolve customer issues in real time.
- **Lifecycle States:**
  OPEN → IN_PROGRESS → RESOLVED → CLOSED
- **Priority Classification:**
  LOW • MEDIUM • HIGH • URGENT
- **Live Filter & Search:**
  Instant title, description, and assignee query filtering with status counters.
- **Design System:**
  Modern YouTube Dark Theme (Black & Blue palette, squarish cards, responsive layout).
- **DevOps Focus:**
  Minimal, clean application core designed for automated delivery, testing, and observability.

Three-Tier Architecture

- **Frontend Tier:**
  React 18 SPA served via unprivileged Nginx reverse proxy (non-root appuser).
- **Backend Tier:**
  FastAPI async REST API with SQLAlchemy 2, running on Uvicorn (UID 10001).
- **Database Tier:**
  PostgreSQL 16 relational database with stateful storage and Alembic migrations.
- **Health & Readiness:**
  /health (liveness check) and /ready (database connectivity probe).
- **Observability:**
  /metrics endpoint scraped by Prometheus Instrumentator.
- **Test Coverage:**
  10/10 passing pytest unit tests on isolated SQLite test database.

MODULE 3 • TOOLING ACROSS ALL 6 DOMAINS

Comprehensive Tech Stack

Application Layer

- Python 3.12 & FastAPI 0.116+
- SQLAlchemy 2.0 & Alembic
- React 18 & Vite SPA
- PostgreSQL 16 Database

Containers & Runtime

- Multi-stage Dockerfiles
- Unprivileged appuser (UID 10001)
- Docker Compose v2 stack
- Named volume persistence

CI/CD & Security

- GitHub Actions (3-job workflow)
- GitHub Container Registry (GHCR)
- Aqua Trivy security scanner
- Git commit SHA image tagging

Infrastructure as Code

- Terraform AWS Provider (v5)
- AWS VPC (4 subnets + NAT)
- AWS EKS v1.31 Control Plane
- Managed EC2 node groups

Kubernetes & Helm

- Helm v3 parameterized chart
- Namespace isolation (helpdesk)
- Ingress routing (/ and /api)
- Horizontal Pod Autoscaler (HPA)

Observability & Metrics

- Prometheus FastAPI Instrumentator
- CoreOS ServiceMonitor CRD
- kube-prometheus-stack
- Custom Grafana RED dashboard

MODULE 4 & 8 • GIT TO CLOUD KUBERNETES PIPELINE

End-to-End System Pipeline



Local Environment (Docker Compose)

- Nginx Frontend: Reverse proxy on port 3001/3000 routing /api to backend.
- FastAPI Backend: Uvicorn service exposing REST API on port 8000.
- PostgreSQL 16: Container with persistent volume devopsprojects_postgres_data.
- Network Isolation: Bridge network connecting services with internal DNS resolution.

Production Cloud (AWS VPC & EKS)

- AWS VPC (ap-south-1): 2 Public subnets for Ingress + 2 Private subnets for compute.
- EKS Cluster v1.31: Managed Kubernetes with auto-scaling t3.small EC2 node groups.
- Helm Deployment: Declarative chart managing pods, services, secrets, and HPA.
- Observability Stack: kube-prometheus-stack scraping /metrics via ServiceMonitor CRD.

MODULE 5 • AUTOMATED 3-JOB DELIVERY WORKFLOW

GitHub Actions CI/CD Pipeline

Job 1: Test

Pytest unit tests (10/10)
React Vite build check
Duration: ~25s • Isolated DB

Job 2: Build & Scan

Multi-stage Docker builds
Aqua Trivy CVE gate
Push images to GHCR (~1m 20s)

Job 3: Deploy

EKS cluster connectivity probe
Helm release upgrade
Live rollout verification (~10s)

Automated Quality Gates

- Pre-Commit Testing: 10 pytest test cases validating CRUD, stats, health, and readiness.
- Zero-Tolerance Security: Aqua Trivy scans images; builds immediately fail on HIGH or CRITICAL CVEs.
- Deterministic Enforcement: Defective or vulnerable code is rejected before reaching container registry.
- Pipeline Status: 100% Green CI execution in GitHub Actions (Run ID: 38847560510).

Commit SHA Image Tagging

- Complete Provenance: Every container running in Kubernetes maps directly to an immutable Git commit hash.
- Eliminates ':latest' Drift: Guarantees identical, predictable container deployments across all cluster nodes.
- Instant Rollbacks: Reverting to any past release requires only a Helm image tag update without rebuilding.
- Published Artifacts: Images published to ghcr.io/yashwardhan-ed/devops-project-backend and frontend.

MODULE 6 • SHIFT-LEFT DEVSECOPS

Trivy Vulnerability Scanning & Fixes

Shift-Left Security Gate: Trivy scans container images in CI; builds fail automatically on HIGH or CRITICAL CVEs.

Vulnerabilities Detected in CI

- **Starlette 0.41.3 (HIGH)**
  CVE-2025-62727: Range header DoS vulnerability
- **AnyIO 4.7.0 (CRITICAL)**
  CVE-2026-63374: TLS certificate domain spoofing flaw
- **Root Execution Risk**
  Default UID 0 execution susceptible to container breakout
- **Nginx Alpine Base**
  Scanned upstream packages for outdated OS libraries

Engineering Fixes Applied

- ✓ **Upgraded AnyIO >= 4.14.2**
  Eliminated CRITICAL TLS spoofing vulnerability
- ✓ **Upgraded Starlette >= 0.49.1**
  Patched Range DoS; preserved Prometheus support
- ✓ **Non-Root Security Context**
  Enforced unprivileged appuser (UID 10001) in Dockerfile
- ✓ **Clean CI Verification Gate**
  0 HIGH / CRITICAL CVEs; pipeline green & images pushed

MODULE 7 • INFRASTRUCTURE AS CODE ON AWS

Terraform VPC & EKS in ap-south-1

AWS Cloud Resources Provisioned

- **VPC & CIDR:**
  Dedicated 10.20.0.0/16 VPC in AWS ap-south-1 (Mumbai).
- **Subnet Isolation:**
  2 Public subnets (Ingress/NAT) + 2 Private subnets (Compute).
- **NAT Gateway:**
  Outbound internet egress for private nodes without public exposure.
- **EKS Cluster v1.31:**
  Managed Kubernetes control plane (helpdesk-eks).
- **Worker Node Group:**
  Auto-scaling group of t3.small EC2 instances (2 to 4 nodes).

Terraform Workflow & Best Practices

- **Automated Lifecycle:**
  Standard init → plan → apply → destroy execution flow.
- **Zero Committed Secrets:**
  *.tfstate and *.tfvars strictly excluded via .gitignore.
- **Template Hygiene:**
  Credentials-free terraform.tfvars.example committed for operators.
- **Cost-Safe Teardown:**
  1-command terraform destroy eliminates orphan cloud costs.

MODULE 8 • ORCHESTRATION, HELM CHART & HPA

Kubernetes, Helm & Autoscaling

Helm Chart (helpdesk)

- **Parameterized Chart**
  Standardized v1.0.0 Helm package
- **Templates**
  Deployments, Services, Ingress, HPA
- **Multi-Environment**
  values.yaml and values-dev.yaml
- **1-Command Deploy**
  helm upgrade --install helpdesk ...

Networking & Ingress

- **Single Entrypoint**
  helpdesk.local domain routing
- **Frontend Path /**
  Routes to React Nginx on port 80
- **Backend Path /api**
  Routes to FastAPI Uvicorn on 8000
- **DNS Aliases**
  ClusterIP services for backend & postgres

Autoscaling (HPA)

- **Scaling Threshold**
  60% target CPU utilization
- **Replica Range**
  2 minimum to 6 maximum pods
- **Stress Tested**
  Verified with 1,000 concurrent requests
- **Zero Downtime**
  Rolling updates with maxSurge=1

MODULE 9 • OBSERVABILITY & RED METRICS

Prometheus Metrics & Grafana



RED Metrics Monitored in Grafana

- **Rate (Throughput):**
  Query: rate(http_requests_total[1m]) | Measures requests per second across all API endpoints.
- **Errors (Reliability):**
  Query: rate(http_requests_total{status=~"5.."}[1m]) | Immediate visual indicator for 5xx unhandled exceptions.
- **Duration (Latency):**
  Query: histogram_quantile(0.95, rate(...)) | p95 latency monitoring ensuring API responses complete in <50ms.
- **Replica Health:**
  Query: Scrapes all active pods | Discovers healthy backend endpoints dynamically via Kubernetes labels.

MODULE 10 • POST-IMPLEMENTATION REVIEW

Key Engineering Insights

Automated Quality Gates

Pre-commit pytest unit tests and Aqua Trivy container scans stop broken builds and zero-day vulnerabilities before registry publishing.

Immutable Commit SHA Tagging

Pinning container tags to Git commit hashes ensures 100% provenance, prevents ':latest' drift, and enables instant rollbacks.

Controlled UI State Management

Replaced fragile synthetic event calls with controlled React state, eliminating async unmount exceptions and ensuring smooth updates.

Cross-Environment DNS Aliases

Configured dual Kubernetes service names (helpdesk-backend and backend alias) to ensure seamless Nginx routing across Compose and K8s.